

sted Document Management System with Secure File Sharing and Versi

- Final Presentation
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Problem Statement

- Traditional file handling often relies on local storage, email attachments, or unmanaged shared folders.
- These methods can lead to duplication, unauthorized access, accidental overwrites, and poor traceability.
- A centralized cloud-based system is needed for secure and reliable document management.

Project Objectives

- Design a structured metadata schema for documents.
- Develop secure file upload and retrieval modules.
- Integrate cloud object storage and enable version control.
- Implement encryption, access permissions, monitoring, and large file support.

Proposed Solution

- A cloud-hosted document management system with centralized storage and controlled access.
- The system supports secure upload, retrieval, sharing, version tracking, and storage monitoring.
- The design focuses on scalability, security, and high availability.

System Architecture

- Client interface for document operations.
- Application server for business logic and request validation.
- Metadata layer for indexing and document records.
- Cloud object storage for centralized file persistence.
- Security, versioning, and monitoring components.

Metadata Schema

- Core fields include document ID, file name, type, size, owner, version, checksum, and storage path.

- Business fields may include category, tags, department, retention class, and confidentiality level.
- Metadata supports search, governance, audit reference, and access mapping.

Security Features

- Encryption in transit using HTTPS or TLS.
- Encryption at rest for stored files.
- Role-based access control for admin, uploader, editor, and viewer.
- Audit logging for access and permission changes.

Version Control and Storage

- Cloud object storage is used for scalable and centralized file management.
- Each updated document creates a new version with version number and timestamp.
- Previous versions can be retained for rollback and traceability.

Testing

- Functional testing for upload, retrieval, permissions, and version creation.

- Performance testing for large file upload and download.
- Validation of integrity, retry behavior, and secure access restrictions.

Conclusion

- The project provides a secure and scalable approach to document management in the cloud.
- It improves storage reliability, access control, and version traceability.
- It also creates a foundation for future enhancements such as OCR, search, and workflow automation.

Links

- GitHub Link: Add your repository URL here.
- Live Demo Link: Add your deployed application URL here.